The D. G. Ruparel College
OF ARTS, SCIENCE AND COMMERCE
Department of Computer Science • Information Technology

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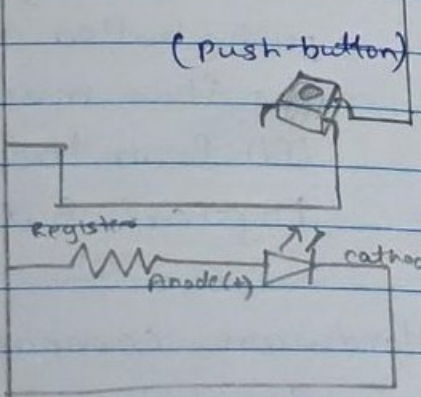
TITLE : Design and implement a Node-RED flow on Raspberry Pi that integrates a physical push button and LED with a web dashboard.

- Aim : To design and implement a Node-RED flow on Raspberry pi that integrates a physical push-button and LED with a web dashboard. The flow must allow the users to toggle the LED from the dashboard and via the Physical button.

- Hardware component Required :
  -Raspberry Pi 400 (Pi version 4), power Adapter, HDMI to micro HDMI cable, mouse, monitor, push button, resistor, LED, Jumper wires.

- circuit Schematic:

Alternate function					Alternate function
	3.3V PWR	1	2	5V PWR	
I2C1 SDA	GPIO 2	3	4	5V PWR	
I2C1 SCL	GPIO 3	5	6	GND	
	GPIO 4	7	8	UART0 Tx	
	GND	9	10	UART0 Rx	
	GPIO 17	11	12	GPIO 18	
	GPIO 27	13	14	GND	
	GPIO 22	15	16	GPIO 23	
	3.3V PWR	17	18	GPIO 24	
SPI0 MOSI	GPIO 10	19	20	GND	
SPI0 MISO	GPIO 9	21	22	GPIO 25	
SPI0 SCLK	GPIO 11	23	24	GPIO 8	SPI0 CS0
	GND	25	26	GPIO 7	SPI0 CS1
	Reserved	27	28	Reserved	
	GPIO 5	29	30	GND	
	GPIO 6	31	32	GPIO 12	
	GPIO 13	33	34	GND	
SPI1 MISO	GPIO 19	35	36	GPIO 16	SPI1 CS0
	GPIO 26	37	38	GPIO 20	SPI1 MOSI
	GND	39	40	GPIO 21	SPI1 SCLK



Anode(+) = longer terminal

cathod(-) = shorter terminal.

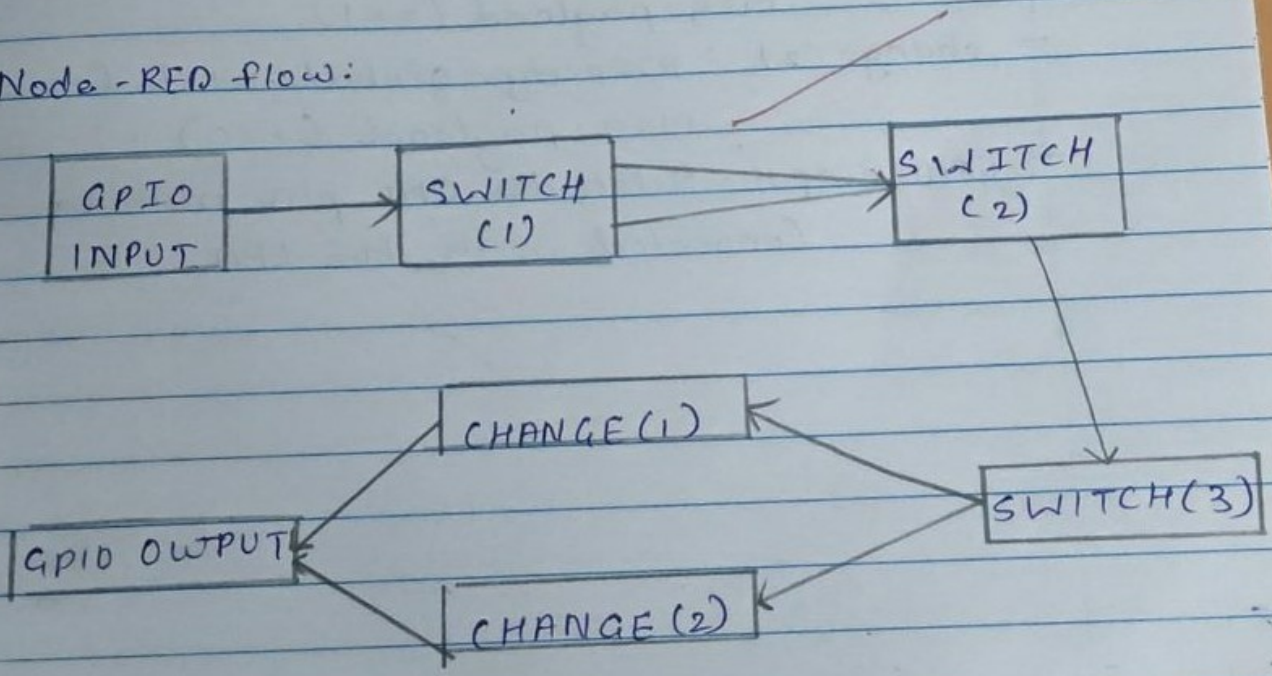
- command:

— bash < (curl -sL https://github.com/node-red/linux-installers/releases/latest/download/install-update-noderd-deb)
[To install Node RED on Raspberry Pi]

— node-red-start [This starts the Node-RED service. open the url displayed on terminal)

— node-red-stop [This stop the Node-RED Service]
— node-red-restart [This restart the Node-RED service]
— node-red-reload [This stops then starts the Node-RED Services]

- Node-RED flow:



- Node Properties:

- GPIO input : select the GPIO pin number connected with the push button.
- switch (1): property - global.state
  Rule 1 - == 0
  Rule 2 - == 1
- switch (2) : Property - msg.payload
  Rule - == 1
- Switch (3) : Property - global.state
  Rule 1 - == 0
  Rule 2 - == 4
- change (1) : Property - global.state (== 1)
  msg.payload (== 1)
- change (2) : property - global.state (== 0)
  msg.payload (== 0)
- GPIO output : Select the GPIO pin number connected with the LED.

Output:

A Node-RED flow was successfully designed and implemented on the Raspberry Pi to control an LED using a physical push button and a web dashboard. The Switch nodes check the stored LED state and iggore the button release (payload not equal to 1), while the change nodes update global. state and msg.payload to turn the LED ON or off. This practical shows how Node-RED allows GPIO hardware to be controlled and monitored through a visual flow-based programming approach, without writing code.